

Bar Graphs

When the data are qualitative or categorical, bar graphs can be used to represent the data. A bar graph can be drawn using either horizontal or vertical bars.

A **bar graph** represents the data by using vertical or horizontal bars whose heights or lengths represent the frequencies of the data.

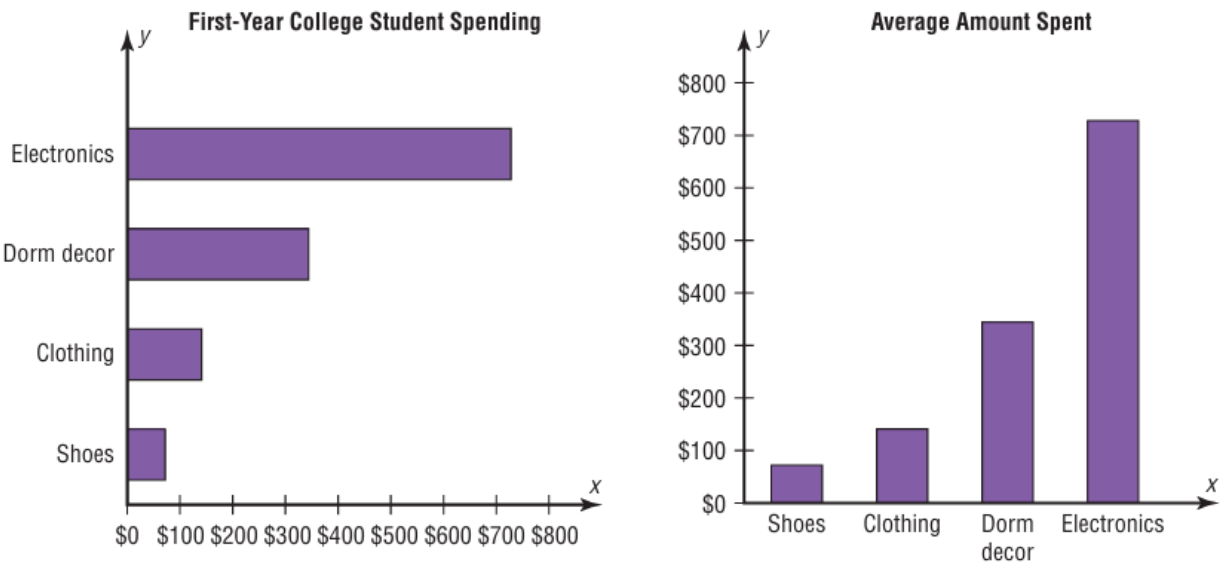
EXAMPLE 2-8 College Spending for First-Year Students

The table shows the average money spent by first-year college students. Draw a horizontal and vertical bar graph for the data.

Electronics	\$728
Dorm decor	344
Clothing	141
Shoes	72

Source: The National Retail Federation.

FIGURE 2-9 Bar Graphs for Example 2-8



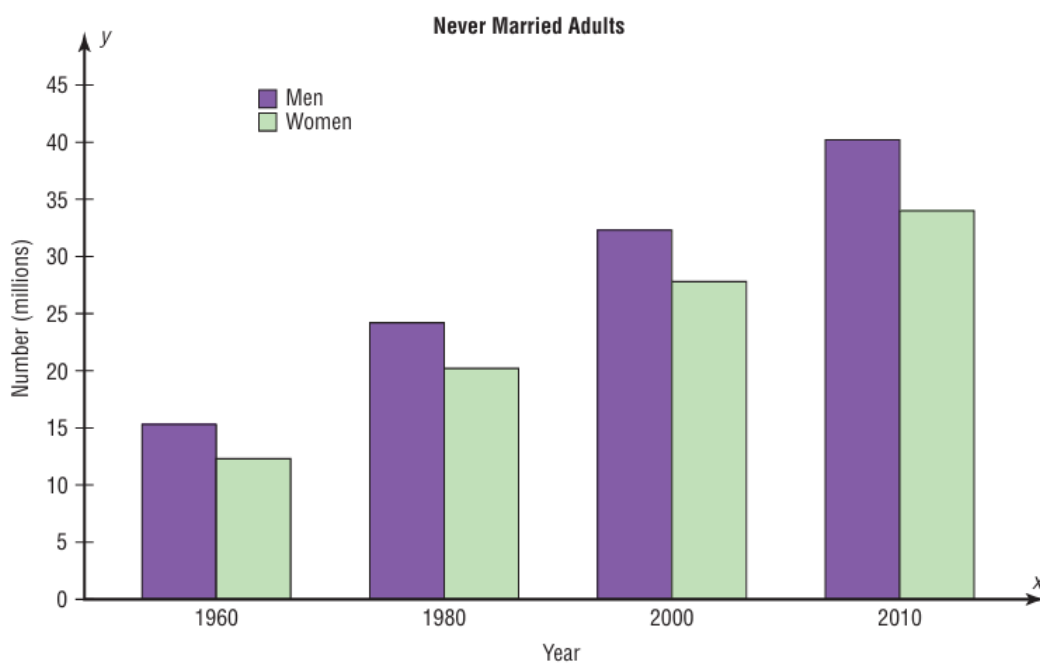
The graphs show that first-year college students spend the most on electronic equipment.

Bar graphs can also be used to compare data for two or more groups. These types of bar graphs are called *compound bar graphs*. Consider the following data for the number (in millions) of never married adults in the United States.

Year	Males	Females
1960	15.3	12.3
1980	24.2	20.2
2000	32.3	27.8
2010	40.2	34.0

Source: U.S. Census Bureau.

Figure 2–10 shows a bar graph that compares the number of never married males with the number of never married females for the years shown. The comparison is made by placing the bars next to each other for the specific years. The heights of the bars can be compared. This graph shows that there have consistently been more never married males



than never married females and that the difference in the two groups has increased slightly over the last 50 years.

Pareto Charts

When the variable displayed on the horizontal axis is qualitative or categorical, a *Pareto chart* can also be used to represent the data.

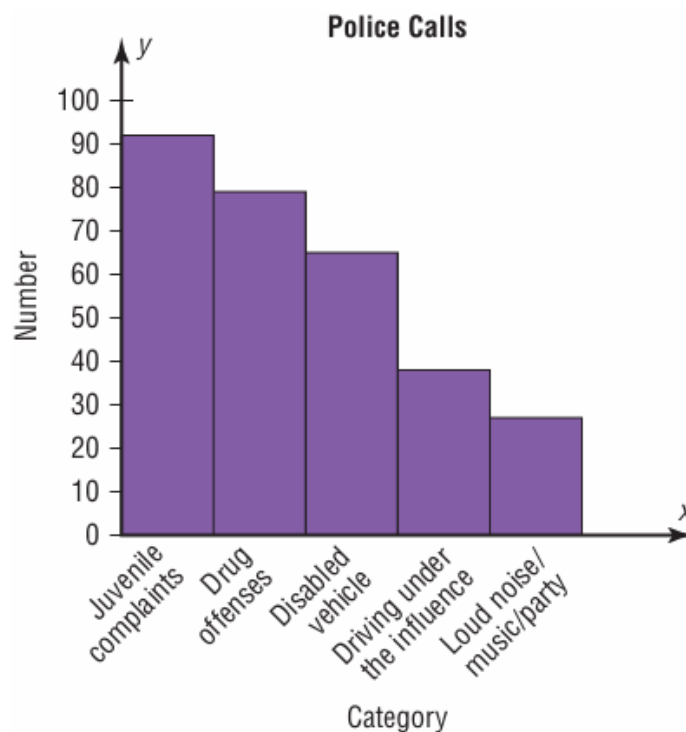
A **Pareto chart** is used to represent a frequency distribution for a categorical variable, and the frequencies are displayed by the heights of vertical bars, which are arranged in order from highest to lowest.

EXAMPLE 2-9 Police Calls

The data shown here consist of the number of police calls for specific categories that a local municipality received in 2011. (The data were obtained by the author.) Draw and analyze a Pareto chart for the data.

Category	Number
Juvenile complaint	92
Loud noise/music/party	27
Drug offenses	79
Driving under the influence	38
Disabled vehicle	65

FIGURE 2-11 Pareto Chart for Example 2-9



Suggestions for Drawing Pareto Charts

1. Make the bars the same width.
2. Arrange the data from largest to smallest according to frequency.
3. Make the units that are used for the frequency equal in size.

The Time Series Graph

When data are collected over a period of time, they can be represented by a time series graph.

A **time series graph** represents data that occur over a specific period of time.

Example 2–10 shows the procedure for constructing a time series graph.

EXAMPLE 2–10 Percent of Cigarette Smokers

The data show the percentage of U.S. adults who smoke. Draw and analyze a time series graph for the data.

Year	1970	1980	1990	2000	2010
Percent	37	33	25	23	19

Source: Center for Disease Control and Prevention.

Historical Note

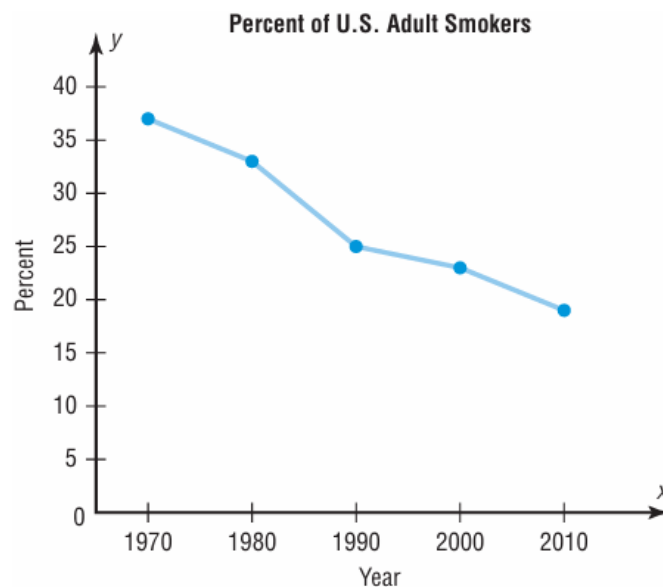
Time series graphs are over 1000 years old. The first ones were used to chart the movements of the planets and the sun.

SOLUTION

- Step 1** Draw and label the x and y axes.
- Step 2** Label the x axis for years and the y axis for the percent.
- Step 3** Plot each point according to the table.
- Step 4** Draw line segments connecting adjacent points. Do not try to fit a smooth curve through the data points. See Figure 2–12.

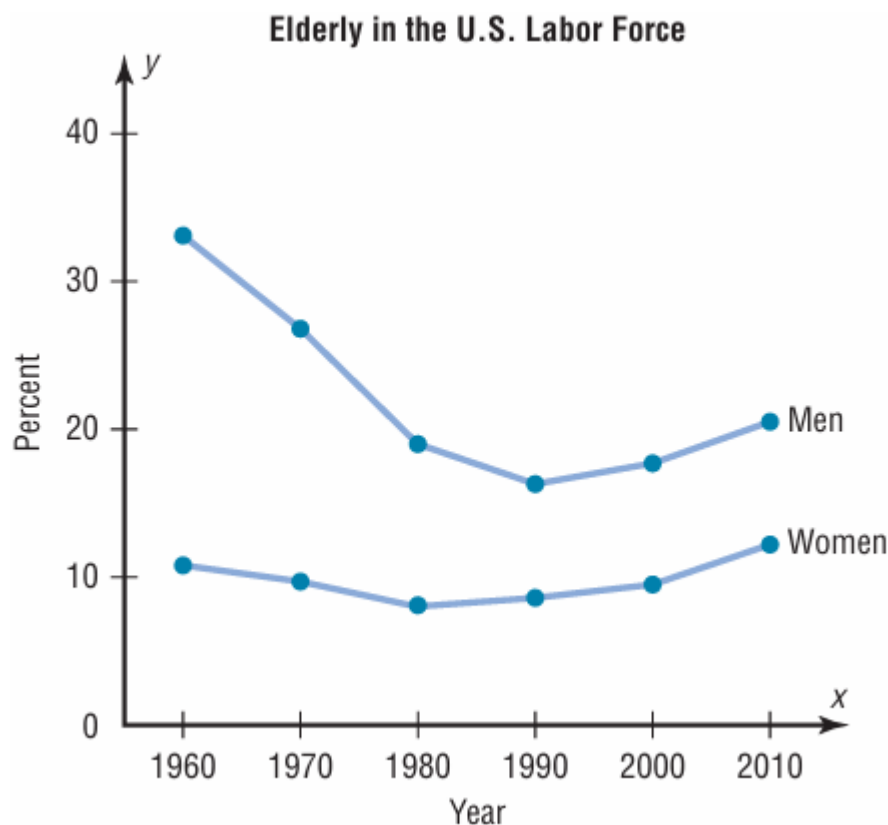
There has been a decrease in the percent of smokers over the 40-year period.

Figure for Example 2–10



When you analyze a time series graph, look for a trend or pattern that occurs over the time period. For example, is the line ascending (indicating an increase over time) or descending (indicating a decrease over time)? Another thing to look for is the slope, or steepness, of the line. A line that is steep over a specific time period indicates a rapid increase or decrease over that period.

Two or more data sets can be compared on the same graph called a *compound time series graph* if two or more lines are used, as shown in Figure 2–13. This graph shows the



percentage of elderly males and females in the U.S. labor force from 1960 to 2010. It shows that the percentage of elderly men decreased significantly from 1960 to 1990 and then increased slightly after that. For the elderly females, the percentage decreased slightly from 1960 to 1980 and then increased from 1980 to 2010.

The Pie Graph

Pie graphs are used extensively in statistics. The purpose of the pie graph is to show the relationship of the parts to the whole by visually comparing the sizes of the sections. Percentages or proportions can be used. The variable is nominal or categorical.

A **pie graph** is a circle that is divided into sections or wedges according to the percentage of frequencies in each category of the distribution.

EXAMPLE 2-11 Super Bowl Snack Foods

This frequency distribution shows the number of pounds of each snack food eaten during the Super Bowl. Construct a pie graph for the data.

Snack	Pounds (frequency)
Potato chips	11.2 million
Tortilla chips	8.2 million
Pretzels	4.3 million
Popcorn	3.8 million
Snack nuts	2.5 million
	Total $n = 30.0$ million

Source: USA TODAY Weekend.

SOLUTION

Step 1 Since there are 360° in a circle, the frequency for each class must be converted to a proportional part of the circle. This conversion is done by using the formula

$$\text{Degrees} = \frac{f}{n} \cdot 360^\circ$$

where f = frequency for each class and n = sum of the frequencies. Hence, the following conversions are obtained. The degrees should sum to 360° .¹

Potato chips	$\frac{11.2}{30} \cdot 360^\circ = 134^\circ$
Tortilla chips	$\frac{8.2}{30} \cdot 360^\circ = 98^\circ$
Pretzels	$\frac{4.3}{30} \cdot 360^\circ = 52^\circ$
Popcorn	$\frac{3.8}{30} \cdot 360^\circ = 46^\circ$
Snack nuts	$\frac{2.5}{30} \cdot 360^\circ = 30^\circ$
Total	$\underline{\quad 360^\circ}$

Step 2 Each frequency must also be converted to a percentage. Recall from Example 2–1 that this conversion is done by using the formula

$$\% = \frac{f}{n} \cdot 100$$

Hence, the following percentages are obtained. The percentages should sum to 100%.²

$$\text{Potato chips} \quad \frac{11.2}{30} \cdot 100 = 37.3\%$$

$$\text{Tortilla chips} \quad \frac{8.2}{30} \cdot 100 = 27.3\%$$

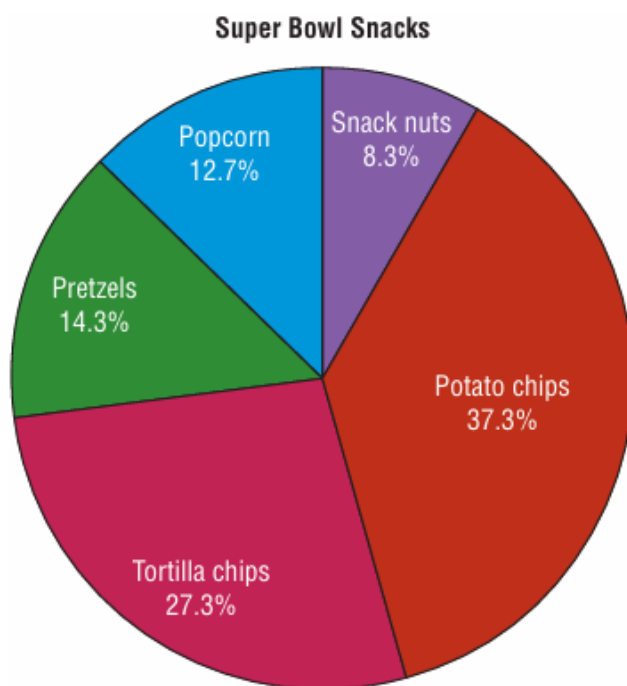
$$\text{Pretzels} \quad \frac{4.3}{30} \cdot 100 = 14.3\%$$

$$\text{Popcorn} \quad \frac{3.8}{30} \cdot 100 = 12.7\%$$

$$\text{Snack nuts} \quad \frac{2.5}{30} \cdot 100 = 8.3\%$$

$$\text{Total} \quad \underline{\quad 99.9\% \quad}$$

FIGURE 2–14 Pie Graph for Example 2–11



Step 3 Next, using a protractor and a compass, draw the graph, using the appropriate degree measures found in Step 1, and label each section with the name and percentages, as shown in Figure 2–14.

Dotplots

A dotplot uses points or dots to represent the data values. If the data values occur more than once, the corresponding points are plotted above one another.

A **dotplot** is a statistical graph in which each data value is plotted as a point (dot) above the horizontal axis.

Dotplots are used to show how the data values are distributed and to see if there are any extremely high or low data values.

EXAMPLE 2-13 Named Storms

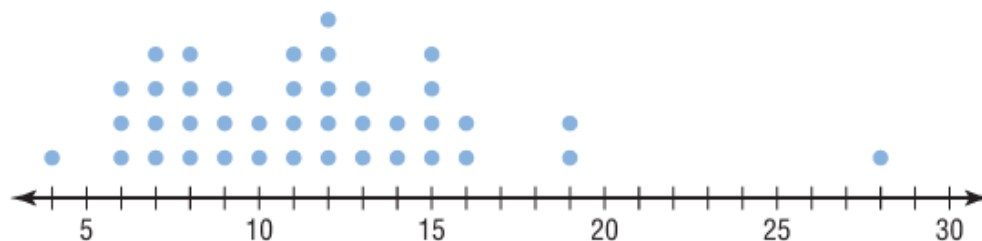
The data show the number of named storms each year for the last 40 years. Construct and analyze a dotplot for the data.

19	15	14	7	6	11	11
9	16	8	8	11	9	8
16	12	13	14	13	12	7
15	15	19	11	4	6	13
10	15	7	12	6	10	
28	12	8	7	12	9	

Source: NOAA.

- Step 1** Find the lowest and highest data values, and decide what scale to use on the horizontal axis. The lowest data value is 4 and the highest data value is 28, so a scale from 4 to 28 is needed.
- Step 2** Draw a horizontal line, and draw the scale on the line.
- Step 3** Plot each data value above the line. If the value occurs more than once, plot the other point above the first point. See Figure 2-16.

FIGURE 2-16 Figure for Example 2-13



The graph shows that the majority of the named storms occur with frequency between 6 and 16 per year. There are only 3 years when there were 19 or more named storms per year.

Stem and Leaf Plots

The stem and leaf plot is a method of organizing data and is a combination of sorting and graphing. It has the advantage over a grouped frequency distribution of retaining the actual data while showing them in graphical form.

A **stem and leaf plot** is a data plot that uses part of the data value as the stem and part of the data value as the leaf to form groups or classes.

For example, a data value of 34 would have 3 as the stem and 4 as the leaf. A data value of 356 would have 35 as the stem and 6 as the leaf.

EXAMPLE 2-14 Out Patient Cardiograms

At an outpatient testing center, the number of cardiograms performed each day for 20 days is shown. Construct a stem and leaf plot for the data.

25	31	20	32	13
14	43	02	57	23
36	32	33	32	44
32	52	44	51	45

SOLUTION

Step 1 Arrange the data in order:

02, 13, 14, 20, 23, 25, 31, 32, 32, 32,
32, 33, 36, 43, 44, 44, 45, 51, 52, 57

Note: Arranging the data in order is not essential and can be cumbersome when the data set is large; however, it is helpful in constructing a stem and leaf plot. The leaves in the final stem and leaf plot should be arranged in order.

Step 2 Separate the data according to the first digit, as shown.

02 13, 14 20, 23, 25 31, 32, 32, 32, 32, 33, 36
43, 44, 44, 45 51, 52, 57

Step 3 A display can be made by using the leading digit as the *stem* and the trailing digit as the *leaf*. For example, for the value 32, the leading digit, 3, is the stem and the trailing digit, 2, is the leaf. For the value 14, the 1 is the stem and the 4 is the leaf. Now a plot can be constructed as shown in Figure 2–17.

Leading digit (stem)	Trailing digit (leaf)
0	2
1	3 4
2	0 3 5
3	1 2 2 2 2 3 6
4	3 4 4 5
5	1 2 7

FIGURE 2–17

Stem and Leaf Plot for
Example 2–14

0	2
1	3 4
2	0 3 5
3	1 2 2 2 2 3 6
4	3 4 4 5
5	1 2 7

Figure 2–17 shows that the distribution peaks in the center and that there are no gaps in the data. For 7 of the 20 days, the number of patients receiving cardiograms was between 31 and 36. The plot also shows that the testing center treated from a minimum of 2 patients to a maximum of 57 patients in any one day.

If there are no data values in a class, you should write the stem number and leave the leaf row blank. Do not put a zero in the leaf row.

EXAMPLE 2-15 Number of Car Thefts in a Large City

An insurance company researcher conducted a survey on the number of car thefts in a large city for a period of 30 days last summer. The raw data are shown. Construct a stem and leaf plot by using classes 50–54, 55–59, 60–64, 65–69, 70–74, and 75–79.

52	62	51	50	69
58	77	66	53	57
75	56	55	67	73
79	59	68	65	72
57	51	63	69	75
65	53	78	66	55

SOLUTION

Step 1 Arrange the data in order.

50, 51, 51, 52, 53, 53, 55, 55, 56, 57, 57, 58, 59, 62, 63,
65, 65, 66, 66, 67, 68, 69, 69, 72, 73, 75, 75, 77, 78, 79

Step 2 Separate the data according to the classes.

50, 51, 51, 52, 53, 53 55, 55, 56, 57, 57, 58, 59
62, 63 65, 65, 66, 66, 67, 68, 69, 69 72, 73
75, 75, 77, 78, 79

Stem and Leaf Plot for
Example 2-15

Step 3 Plot the data as shown here.

5	0	1	1	2	3	3		
5	5	5	6	7	7	8	9	
6	2	3						
6	5	5	6	6	7	8	9	9
7	2	3						
7	5	5	7	8	9			

Leading digit (stem)	Trailing digit (leaf)
5	0 1 1 2 3 3
5	5 5 6 7 7 8 9
6	2 3
6	5 5 6 6 7 8 9 9
7	2 3
7	5 5 7 8 9

The graph for this plot is shown in Figure 2-18.

When you analyze a stem and leaf plot, look for peaks and gaps in the distribution. See if the distribution is symmetric or skewed. Check the variability of the data by looking at the spread.

Related distributions can be compared by using a back-to-back stem and leaf plot. The back-to-back stem and leaf plot uses the same digits for the stems of both distributions, but the digits that are used for the leaves are arranged in order out from the stems on both sides. Example 2-16 shows a back-to-back stem and leaf plot.

EXAMPLE 2-16 Number of Stories in Tall Buildings

The number of stories in two selected samples of tall buildings in Atlanta and Philadelphia is shown. Construct a back-to-back stem and leaf plot, and compare the distributions.

Atlanta					Philadelphia				
55	70	44	36	40	61	40	38	32	30
63	40	44	34	38	58	40	40	25	30
60	47	52	32	32	54	40	36	30	30
50	53	32	28	31	53	39	36	34	33
52	32	34	32	50	50	38	36	39	32
26	29								

Source: *The World Almanac and Book of Facts*.

SOLUTION

- Step 1** Arrange the data for both data sets in order.
- Step 2** Construct a stem and leaf plot, using the same digits as stems. Place the digits for the leaves for Atlanta on the left side of the stem and the digits for the leaves for Philadelphia on the right side, as shown. See Figure 2-19.

FIGURE 2-19 Back-to-Back Stem and Leaf Plot for Example 2-16

Atlanta		Philadelphia	
	9 8 6	2	5
8 6 4 4 2 2 2 2 2 1	3	0 0 0 0 2 2 3 4 6 6 6 8 8 9 9	
7 4 4 0 0	4	0 0 0 0	
5 3 2 2 0 0	5	0 3 4 8	
3 0	6	1	
0	7		

- Step 3** Compare the distributions. The buildings in Atlanta have a large variation in the number of stories per building. Although both distributions are peaked in the 30- to 39-story class, Philadelphia has more buildings in this class. Atlanta has more buildings that have 40 or more stories than Philadelphia does.

The Five-Number Summary and Boxplots

A **boxplot** can be used to graphically represent the data set. These plots involve five specific values:

1. The lowest value of the data set (i.e., minimum)
2. Q_1
3. The median
4. Q_3
5. The highest value of the data set (i.e., maximum)

These values are called a **five-number summary** of the data set.

A **boxplot** is a graph of a data set obtained by drawing a horizontal line from the minimum data value to Q_1 , drawing a horizontal line from Q_3 to the maximum data value, and drawing a box whose vertical sides pass through Q_1 and Q_3 with a vertical line inside the box passing through the median or Q_2 .

Procedure Table

Constructing a Boxplot

- | | |
|---------------|--|
| Step 1 | Find the five-number summary for the data. |
| Step 2 | Draw a horizontal axis and place the scale on the axis. The scale should start on or below the minimum data value and end on or above the maximum data value. |
| Step 3 | Locate the lowest data value, Q_1 , the median, Q_3 , and the highest data value; then draw a box whose vertical sides go through Q_1 and Q_3 . Draw a vertical line through the median. Finally, draw a line from the minimum data value to the left side of the box, and draw a line from the maximum data value to the right side of the box. |

EXAMPLE 3–37 Number of Meteorites Found

The number of meteorites found in 10 states of the United States is 89, 47, 164, 296, 30, 215, 138, 78, 48, 39. Construct a boxplot for the data.

Source: Natural History Museum.

SOLUTION

Step 1 Find the five-number summary for the data.

Arrange the data in order:

30, 39, 47, 48, 78, 89, 138, 164, 215, 296

Find the median.

30, 39, 47, 48, 78, 89, 138, 164, 215, 296



Median

$$\text{Median} = \frac{78 + 89}{2} = 83.5$$

Find Q_1 .

30, 39, 47, 48, 78



Q_1

Find Q_3 .

89, 138, 164, 215, 296

↑
 Q_3

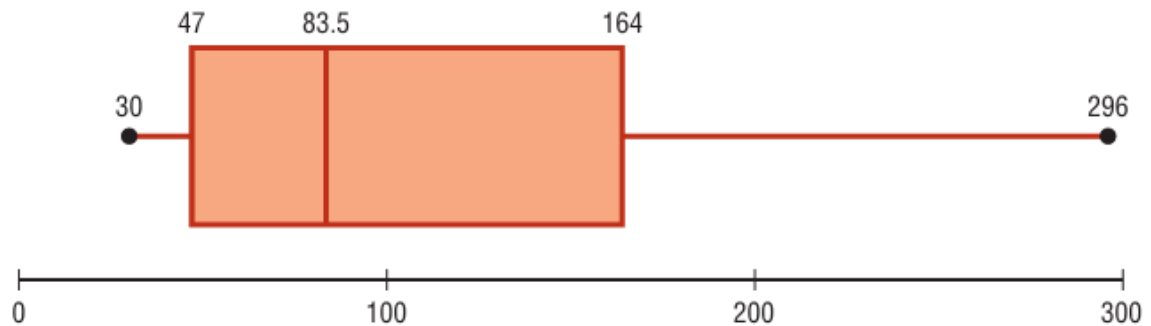
The minimum data value is 30, and the maximum data value is 296.

Step 2 Draw a horizontal axis and the scale.



Step 3 Draw the box above the scale using Q_1 and Q_3 . Draw a vertical line through the median, and draw lines from the lowest data value to the box and from the highest data value to the box. See Figure 3–7.

FIGURE 3–7 Boxplot for Example 3–37



Information Obtained from a Boxplot

1.
 - a. If the median is near the center of the box, the distribution is approximately symmetric.
 - b. If the median falls to the left of the center of the box, the distribution is positively skewed.
 - c. If the median falls to the right of the center, the distribution is negatively skewed.
2.
 - a. If the lines are about the same length, the distribution is approximately symmetric.
 - b. If the right line is larger than the left line, the distribution is positively skewed.
 - c. If the left line is larger than the right line, the distribution is negatively skewed.

The boxplot in Figure 3–7 indicates that the distribution is slightly positively skewed.

If the boxplots for two or more data sets are graphed on the same axis, the distributions can be compared. To compare the averages, use the location of the medians. To compare the variability, use the interquartile range, i.e., the length of the boxes. Example 3–38 shows this procedure.

EXAMPLE 3–38 Sodium Content of Cheese

A dietitian is interested in comparing the sodium content of real cheese with the sodium content of a cheese substitute. The data for two random samples are shown. Compare the distributions, using boxplots.

Real cheese				Cheese substitute			
310	420	45	40	270	180	250	290
220	240	180	90	130	260	340	310

Source: *The Complete Book of Food Counts*.

SOLUTION

Step 1 Find the five-number summary for each data set. For real cheese

40 45 90 180 220 240 310 420

↑

↑

↑

Q_1

MD

Q_3

$$Q_1 = \frac{45 + 90}{2} = 67.5$$

$$MD = \frac{180 + 220}{2} = 200$$

$$Q_3 = \frac{240 + 310}{2} = 275$$

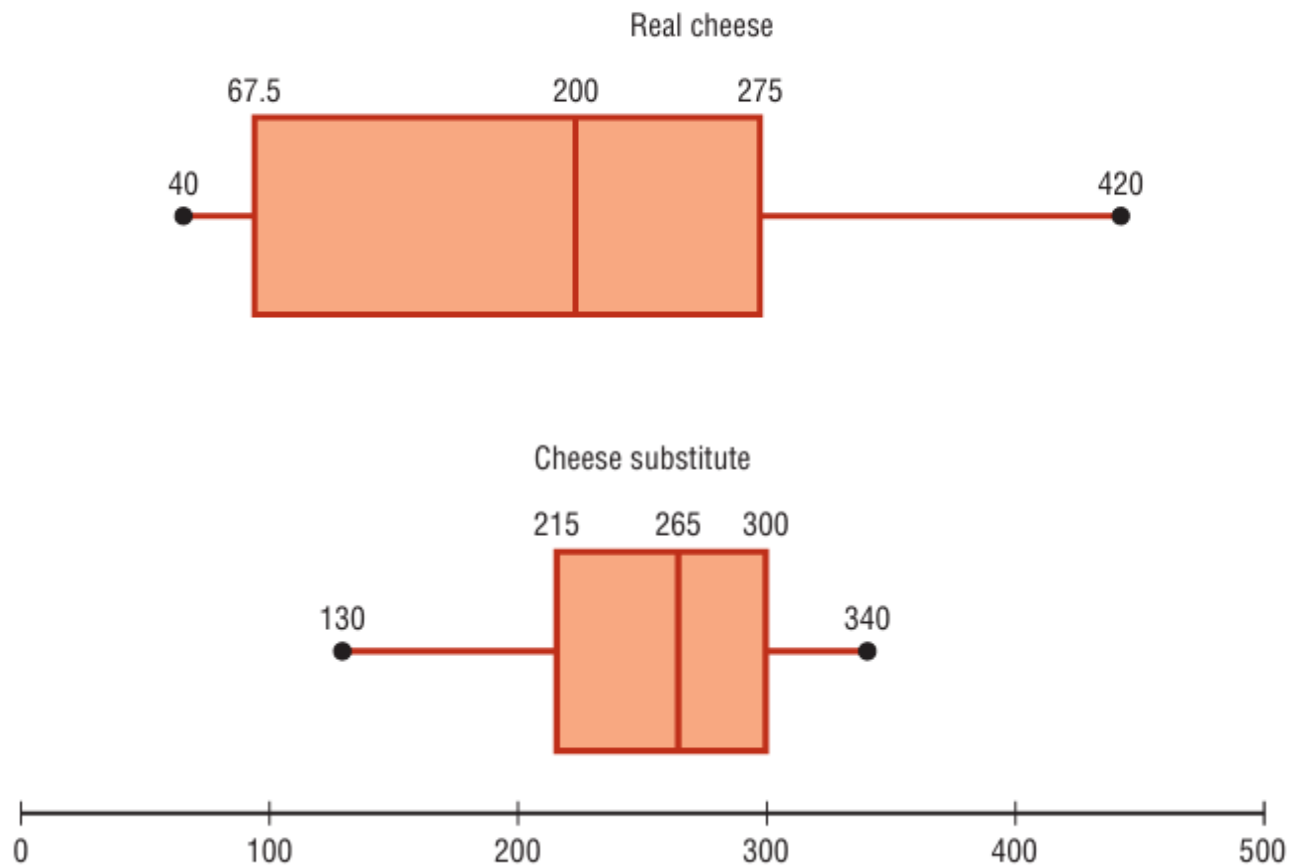
For cheese substitute

130	180	250	260	270	290	310	340
		↑		↑		↑	
		Q_1		MD		Q_3	

$$Q_1 = \frac{180 + 250}{2} = 215 \qquad MD = \frac{260 + 270}{2} = 265$$

$$Q_3 = \frac{290 + 310}{2} = 300$$

FIGURE 3-8 Boxplots for Example 3-38

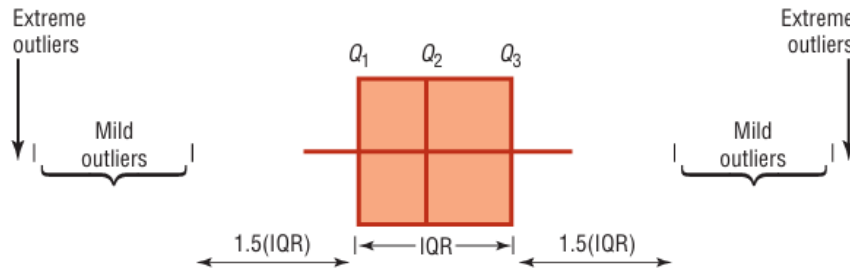


Step 2 Draw the horizontal axis and the scale.

Step 3 Draw the boxplots. See Figure 3-8. Compare the plots. It is quite apparent that the distribution for the cheese substitute data has a higher median than the median for the distribution for the real cheese data. The variation or spread for the distribution of the real cheese data is larger than the variation for the distribution of the cheese substitute data.

Unhealthy Smog Days A *modified boxplot* can be drawn by placing a box around Q_1 and Q_3 and then extending the whiskers to the highest and/or lowest values within 1.5 times the interquartile range

(that is, $Q_3 - Q_1$). *Mild outliers* are values greater than $Q_3 + 1.5(IQR)$ or less than $Q_1 - 1.5(IQR)$. *Extreme outliers* are values greater than $Q_3 + 3(IQR)$ or less than $Q_1 - 3(IQR)$.



For the data shown here, draw a modified boxplot and identify any mild or extreme outliers. The data represent the number of unhealthy smog days for a specific year for the highest 10 locations.

97	39	43	66	91
43	54	42	53	39

Source: U.S. Public Interest Research Group and Clean Air Network.